

CSSS 321

Lab 3: Common Errors

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Today's lab

1. Reading **error messages**. Reference: David Robinson.
2. Getting **help**: reproducible code
3. In-lab exercise

Reading error messages

```
x <- 10 + foo
```

Error: object 'foo' not found

Issue: You tried to access a variable / object that doesn't exist.

You might have:

- **misspelled** the variable name
- incorrectly **capitalized** the variable name (R is case sensitive!)
- **forgotten** to run the line that defines the variable in the first place, or run into an error in that line.

Reading error messages

```
x <- foo(...)
```

Error: could not find function "foo"

Issue: You tried to use a function that doesn't exist. You might have:

- **misspelled** the function name
- incorrectly **capitalized** the function name
- forgotten to **load the library** that provides this function.

Reading error messages

```
x <- c(1:10))
```

Error: unexpected ')' in ...

Issue: There is an extra end parenthesis in your line!

```
x <- 10; y <- 20  
mean(x y)
```

Error: unexpected symbol in ...

Issue: The most common cause of this is forgetting a punctuation mark such as a comma: for example, `foo(bar1 bar2)` instead of `foo(bar1, bar2)`.

Error: unexpected numeric constant is similar: it just means the value after the missing punctuation is a number (for example, `x 2` instead of `x = 2`).

Reading error messages

```
paste("welcome to CSSS, 321)
```

- You might see a + sign in the interpreter after you hit return. This means the previous statement is unfinished:
- it might have an **open parenthesis** that never closes,
 - an open " or ' that is unmatched, or
 - it could end with an operator like + or - that expects the line to continue afterwards.
- Find the problem in your previous lines (count parentheses, and check your quotes) and fix it.

Getting help: reproducible code

- If you feel stuck with an error, or your code does not do what you think it should, you should troubleshoot it with someone else! But, remember to provide **reproducible code** in an R-script file:
 1. Load necessary **packages** at the beginning.
 2. Include all code up to the error, or at least the **necessary** code to reproduce the error you are experiencing.
 3. **Comment** on your code for clarity.
 4. If applicable, send the necessary **data** to reproduce the error.

In-lab exercise.

- Open R-Markdown *W3_subset_logicals.Rmd* and complete the exercise.